

Lizards

PHYLUM	Chordata
CLASS	Reptilia
ORDER	Squamata
SUBORDER	Lacertilia
FAMILIES	37
SPECIES	About 6,300

CLASSIFICATION NOTE

The classification of lizards, like that of snakes, is the subject of much debate, especially the way they are grouped into families. However, their division into the following superfamilies is widely accepted:

- Iguanas and relatives
see pp.410–6
- Geckos and snake-lizards
see pp.417–9
- Skinks and relatives
see pp.420–5
- Anguimorph lizards
see pp.426–30

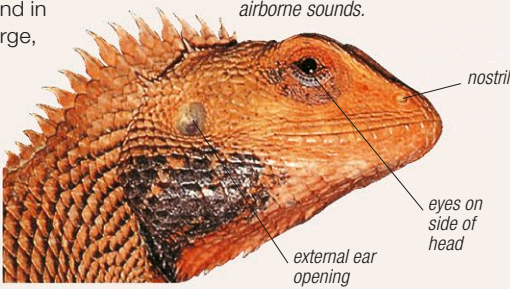
Lizards are the most successful group of reptiles. They form a large and varied group that is not easy to define concisely: although a typical lizard has a distinct head, 4 well-developed limbs, and a long tail, there are exceptions to all of these generalizations. Some lizards also have an unusual ability to shed their tail when attacked by a predator and then regenerate it. Most species reproduce by laying eggs, although some give birth to live young; a small number actively care for their offspring. Lizards have adapted to survive in habitats all over the world (although they are not found in Antarctica). Most live on the ground, on low-lying rocks, or in trees. There is also a small number of burrowing

species and even one that feeds in seawater. A few lizards, notably the geckos, have adapted to live around humans, and in some parts of the world, they are a common sight on the walls and ceilings of buildings.

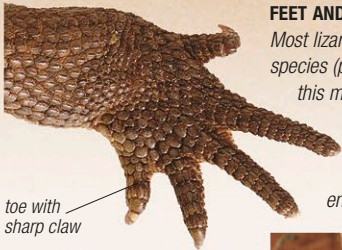
DESERT LIZARDS
Deserts are an important habitat for lizards, such as these web-footed geckos. Like all reptiles, they do not require large amounts of water because they use only small amounts to expel waste. Most of the water they need is obtained from dew and fog. Webbed toes allow them to walk on soft sand.

Anatomy

In addition to a distinct head, 4 legs, and a tail, most lizards have external ear openings, and eyelids that can be opened and closed. However, some lizards have short legs or none at all. There are also forms with no eyes or no ears, while others have eyes covered by a transparent scale rather than movable lids. The tongue may be short or long and is usually notched at the tip; forked tongues are found in some species. A lizard's scales may be large, smooth, and overlapping or small, rough, and studlike, in almost any combination. Many have irregular scales or a few large scales scattered among many smaller ones, but none has the single row of wide ventral scales characteristic of snakes (see p.384). Most lizards have sharp teeth along the edges of their jaws.



SENSES
A lizard's senses are adapted to its lifestyle. Many species that live underground cannot see, while those living above ground may have good vision, but only a few are able to see colors. Lizards with external eardrums can hear airborne sounds.



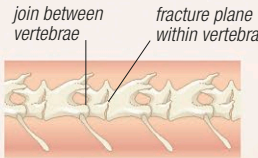
FEET AND CLAWS
Most lizards have 5 digits on each foot. Many species (particularly those that climb, such as this monitor lizard) have sharp claws. Most lizards cannot oppose their digits, although chameleons have their digits fused into 2 opposing groups, enabling them to grip branches.



LEGLESS LIZARDS
Over time, some lizards (such as this Burton's snake-lizard) have lost their limbs. This is an adaptation that helps them to move easily through soil burrows, sand, and leaf litter.

Regeneration

Some lizards are able to voluntarily break off part of their tail, a phenomenon known as autotomy. This is a defense mechanism that helps the lizard escape when grabbed by the tail. To add to the effectiveness of the strategy, the tail may be brightly colored, often blue, to draw the attacker's attention away from the head. The tail may also continue to move for several minutes after being severed, apparently to distract the predator. The tail breaks at one of several predetermined weak points (see below), and it does not have to be pulled in order to cause a break. Most lizards can regrow a broken tail, but the replacement will not have the same scale arrangement or coloration as the original. A lizard may shed its tail several times during its lifetime.



FRACTURE PLANES
A lizard's tail breaks at one of several weak points, called fracture planes, across certain vertebrae. The muscles are arranged in such a way that they, too, will come apart neatly.



NEW GROWTH
This viviparous lizard has shed its tail and grown a replacement. The point at which the old tail was broken can be identified by the change in scale color.